



Enabling Onboard Navigation for UAVs in GPS Denied Environments

MIP400A : B.Tech Project - MidTerm Evaluation

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Goal: UAV self-localization when GPS/GNSS is unavailable, jammed, or spoofed, using only onboard sensors.

- ❑ **This semester:** working end-to-end pipeline, in simulation.
- ❑ **Next semester (through Apr 2027):** fast, power-efficient (FPGA) onboard deployment.

Core approach: Scene-Aided Navigation

Replace GPS with passive measurements the aircraft can already sense: terrain, gravity, magnetic field, and visual scene, matched against a reference map.



Weak signal

GNSS is easily jammed, spoofed, or simply unavailable.

Mechanical view

The IMU already senses acceleration, rotation, and gravity; a camera could replace the satellite link entirely.

Must keep flying

Survey, disaster-response, and defense UAVs need to operate even when GPS fails.

Precedent

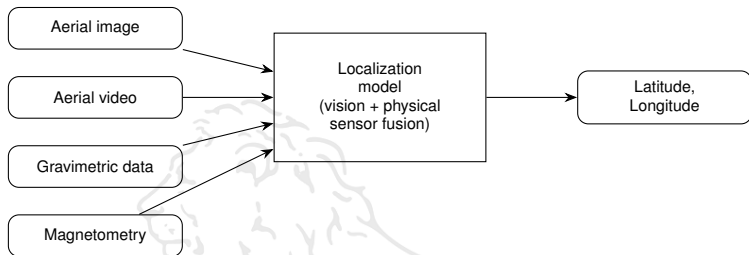
TERCOM and DSMAC proved passive, GPS-free terrain and scene matching decades ago.



Passive GPS-denied navigation methods, screened for feasibility on a small UAV:

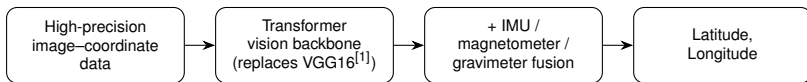
Method	Physical quantity measured	Passive?	Assessment
Terrain matching (TERCOM)	Ground elevation	Yes	Proven, strong candidate ^[4,6]
Visual landmark matching	Camera imagery	Yes	Strong if paired with AI matching ^[1-3,5,7,12,13]
Transformer-based matching	Camera imagery	Yes	SOTA in this niche; motivates our redesign ^[14,15]
Gravity matching	Gravity anomalies	Yes	Promising, needs sensitive sensor ^[8,9]
Gravity-gradient nav.	$\partial g / \partial x, y, z$	Yes	Very promising, higher precision ^[8,9]
Magnetic anomaly nav.	Magnetic field	Yes	Useful in mountainous terrain ^[10,11]
Quantum inertial sensing	Acceleration / rotation	Yes	No map needed; pure mechanics
Wi-Fi / cellular / RF	Radio signal strength	No	Rejected: unavailable at altitude

Reviewed: VGG16^[1], CLIP^[2,3,7], SIFT/ORB^[5], and DEM terrain-weighting^[4,6], achieving sub-3–7 m RMSE^[2,4].



- ❑ **Baseline validated:** replicated the reference implementation^[1] in simulation; accuracy matches the paper's reported numbers.
- ❑ **Now:** redesigning the backbone as a **transformer** (next slide), trading compute and latency for higher accuracy, then adding multimodal sensor fusion.

Proposed Model Architecture



Architecture diagram (to be inserted)

Timeline of the Work



Phase 1: Now to Nov 2026

Transformer backbone +
multimodal fusion
(IMU, magnetometer, gravimeter)
end-to-end pipeline, in simulation

Phase 2: Jan to Apr 2027

Optimize for speed
& power on FPGA
physical UAV test

Phase 2 extends our lab project from last semester, which showed FPGAs are highly power-efficient for this class of computation.



- [1] H. Goforth and S. Lucey, *GPS-Denied UAV Localization using Pre-existing Satellite Imagery*, CMU Robotics Institute.
<https://publications.ri.cmu.edu/gps-denied-uav-localization-using-pre-existing-satelliteimagery>
(code: <https://github.com/hmgoforth/gps-denied-uav-localization>)
- [2] *ngps_flight*: real-time CLIP-style UAV-satellite matching, ~3 m RMSE on Jetson Orin.
https://github.com/snktshrma/ngps_flight
- [3] *CLIP-UAV-localization*: CLIP-based network matching UAV imagery, also recovers heading.
<https://github.com/codebra721/CLIP-UAV-localization>
- [4] *GNSS-Denied-UAV-Geolocalization*: terrain-weighted optimization using satellite maps and DEM data, sub-7 m.
<https://github.com/YFS90/GNSS-Denied-UAV-Geolocalization>
- [5] *VisionUAV-Navigation*: classical SIFT/ORB hybrid feature detection, no deep learning.
<https://github.com/sidharthmohannair/VisionUAV-Navigation>
- [6] GNSS-denied UAV geolocalization via terrain-weighted constraint optimization, sub-7 m error, *ScienceDirect*.
<https://www.sciencedirect.com/science/article/pii/S1569843224006332>
- [7] CLIP-based UAV-satellite image matching, also derives heading direction, *ScienceDirect*.
<https://www.sciencedirect.com/science/article/pii/S2590123025041787>





- [8] A Review of Key Technologies in Gravity Matching Navigation, *Sensors* (2026).
<https://www.mdpi.com/1424-8220/26/13/4208>
- [9] Gravity-Aided Navigation Using Viterbi Map Matching Algorithm. <https://arxiv.org/abs/2204.10492>
- [10] Absolute Positioning Using the Earth's Magnetic Anomaly Field. <https://doi.org/10.1002/navi.138>
- [11] MagNav-Based Navigation Simulation Using Magnetic Anomaly Maps for GPS-Denied Environments.
<https://doi.org/10.5302/J.ICRDS.2026.25.0344>
- [12] *PCVLabDrone2021*: real-time down-facing camera pipeline with quad-tree map retrieval, tested live.
<https://github.com/OSUPCVLab/PCVLabDrone2021>
- [13] *geolocalization*: dual/Siamese network recognizing same-scene UAV and satellite image pairs.
<https://github.com/abhinavtripathi95/geolocalization>
- [14] X. Zhu et al., *TransGeo: Transformer Is All You Need for Cross-view Image Geo-localization*, CVPR 2022.
<https://arxiv.org/abs/2204.00097>
- [15] *FSRA*: A Transformer-Based Feature Segmentation and Region Alignment Method for UAV-View Geo-Localization.
<https://arxiv.org/pdf/2201.09206>

